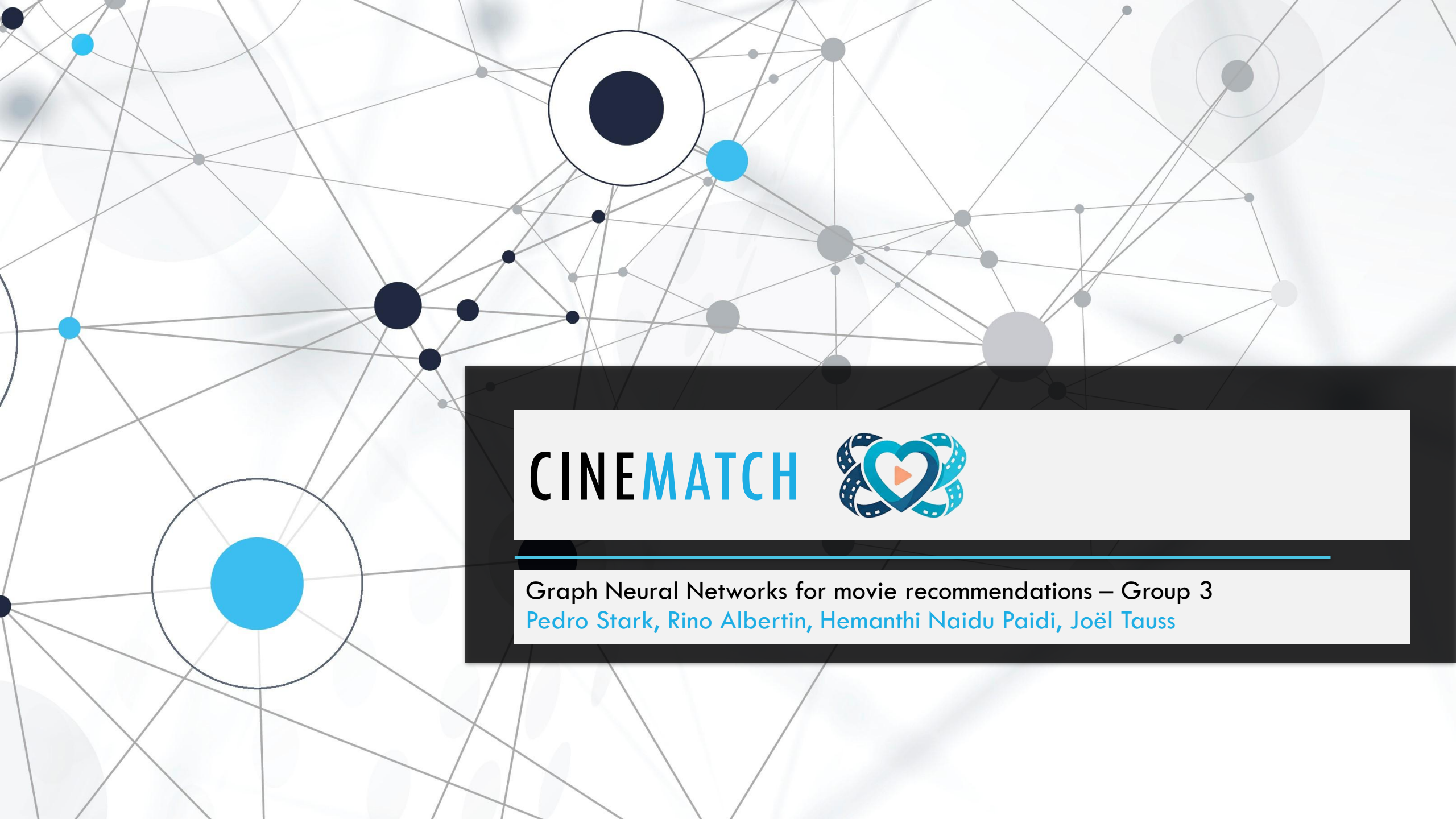




CINEMATCH



Graph Neural Networks for movie recommendations – Group 3
Pedro Stark, Rino Albertin, Hemanthi Naidu Paidi, Joël Tauss

OVERVIEW



Goal:

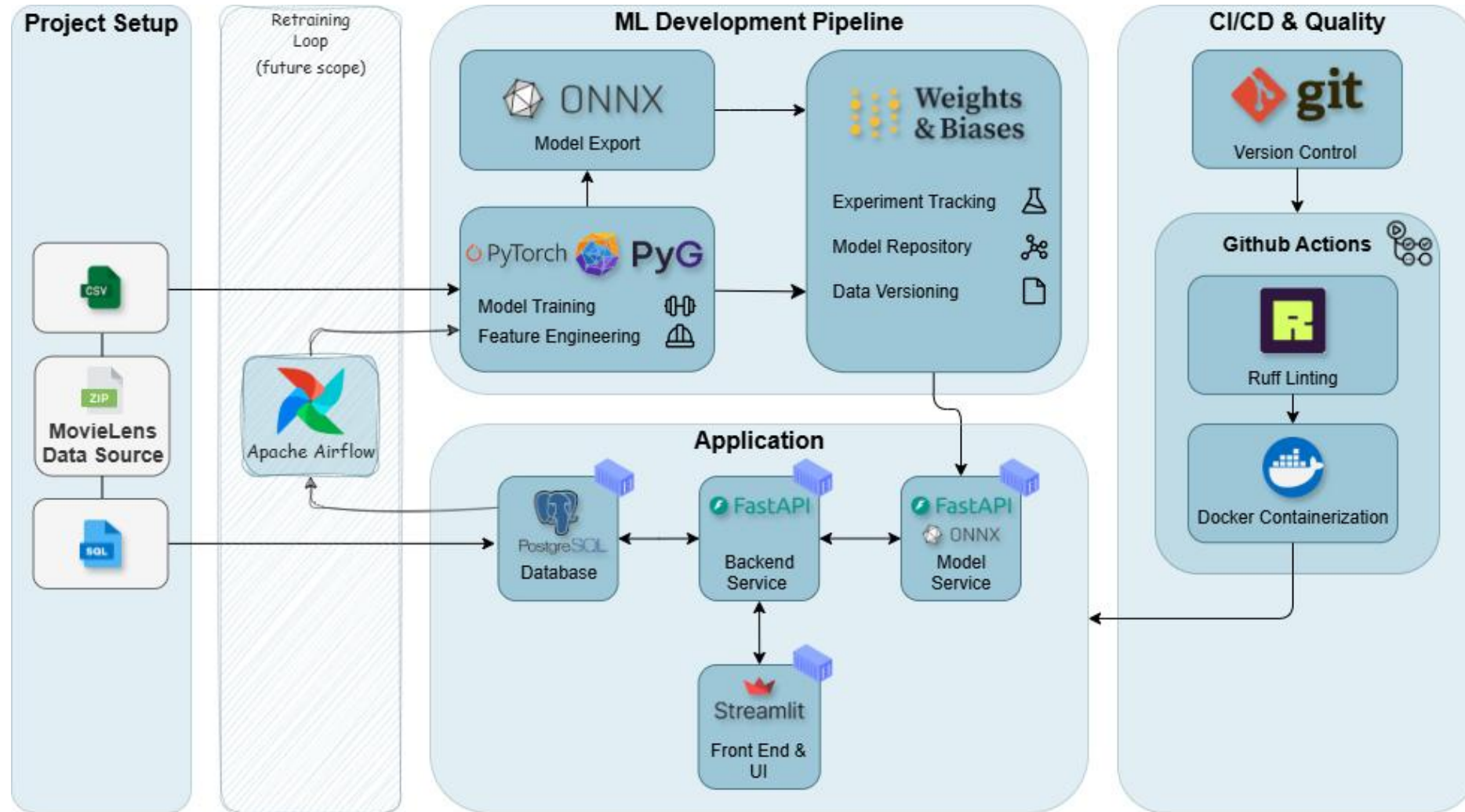
- Help film enjoyers to find new movies
- Recommendations based on personal ratings
- A centralized Platform



Solution:

- A recommendation system based on a Graph neural network
- An easily accessible Website
- (Regularly retrained on new data)

SYSTEM ARCHITECTURE



DATA



- Movies

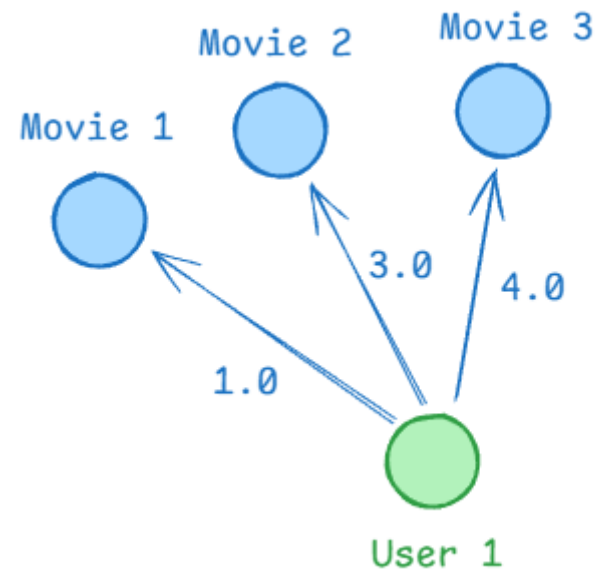
- Genres one-hot encoded
- Embedding for movie title with Sentence Transformer

- Users

- No initial features

- Ratings

- [1..5]

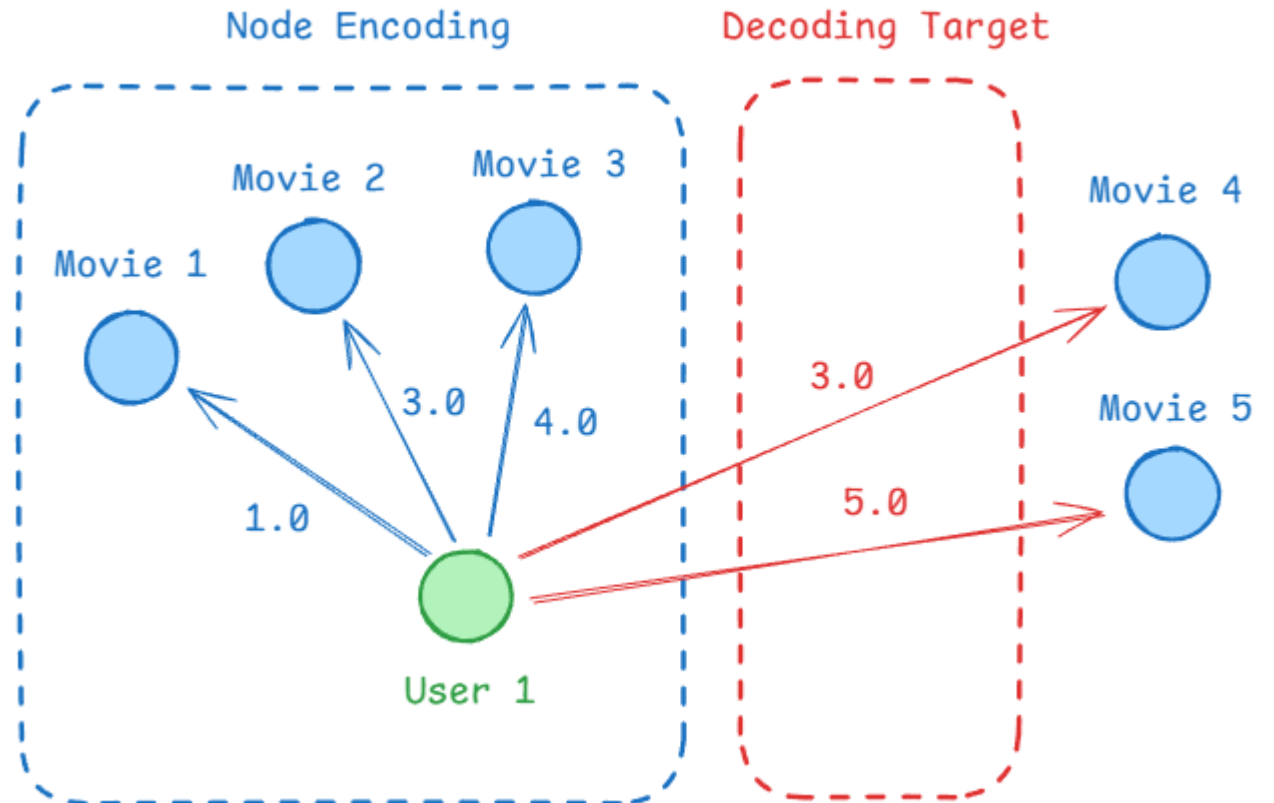


MODEL

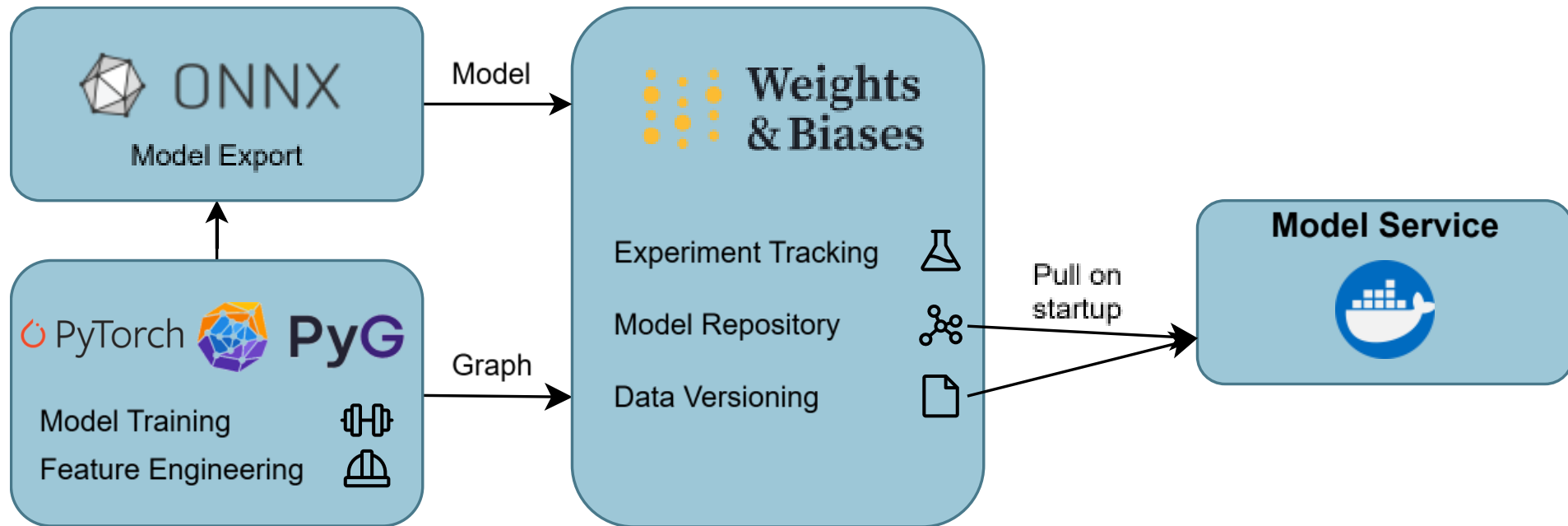


- Node encoder:  **PyG**

- Edge decoder  **PyTorch**



MODEL



BACKEND & DATABASE



FastAPI:

- Fast and easy development
- Provides async functionality
- **Auto documentation with Swagger**



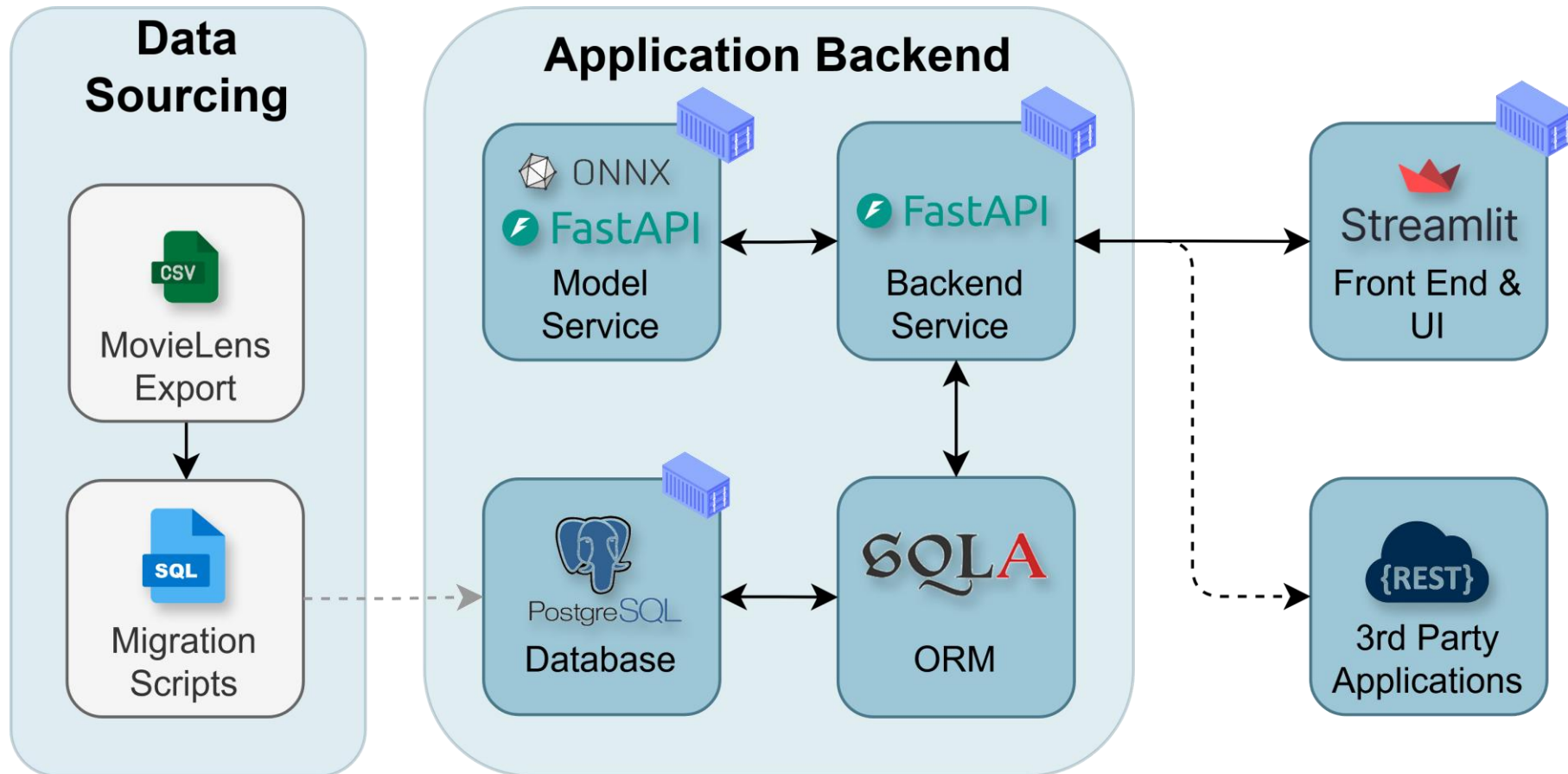
PostgreSQL:

- Open source and large community
- Provides the needed RDBMS functionality
- **Excellent DockerHub support**

Additional Libraries:

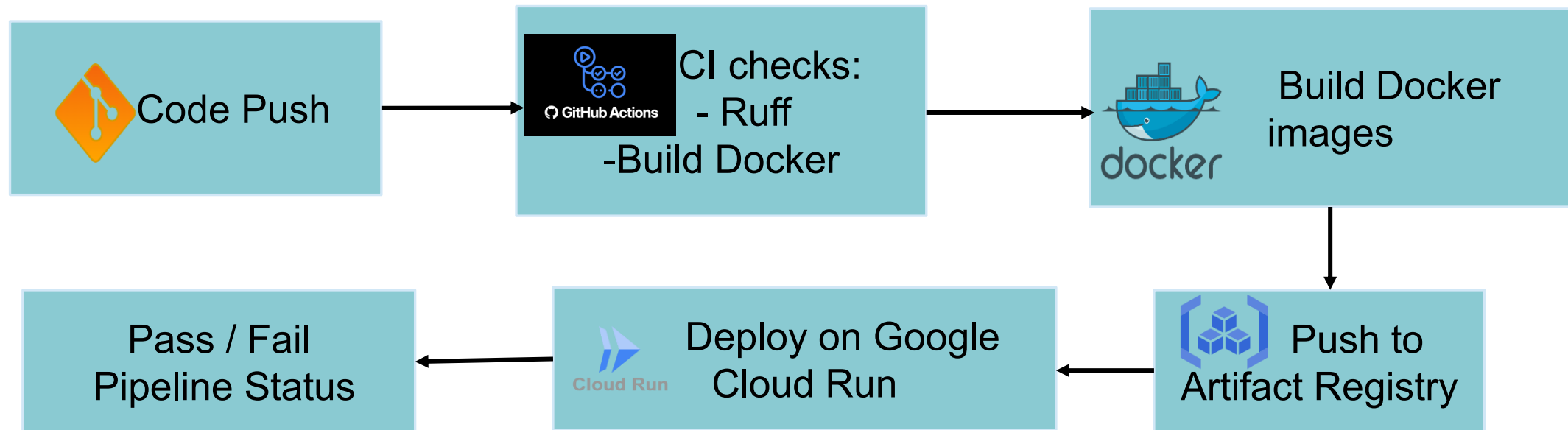
- SQLAlchemy
- Pydantic

BACKEND & DATABASE





CONTINUOUS INTEGRATION AND CONTINUOUS DEPLOYMENT (CI\CD) PIPELINES

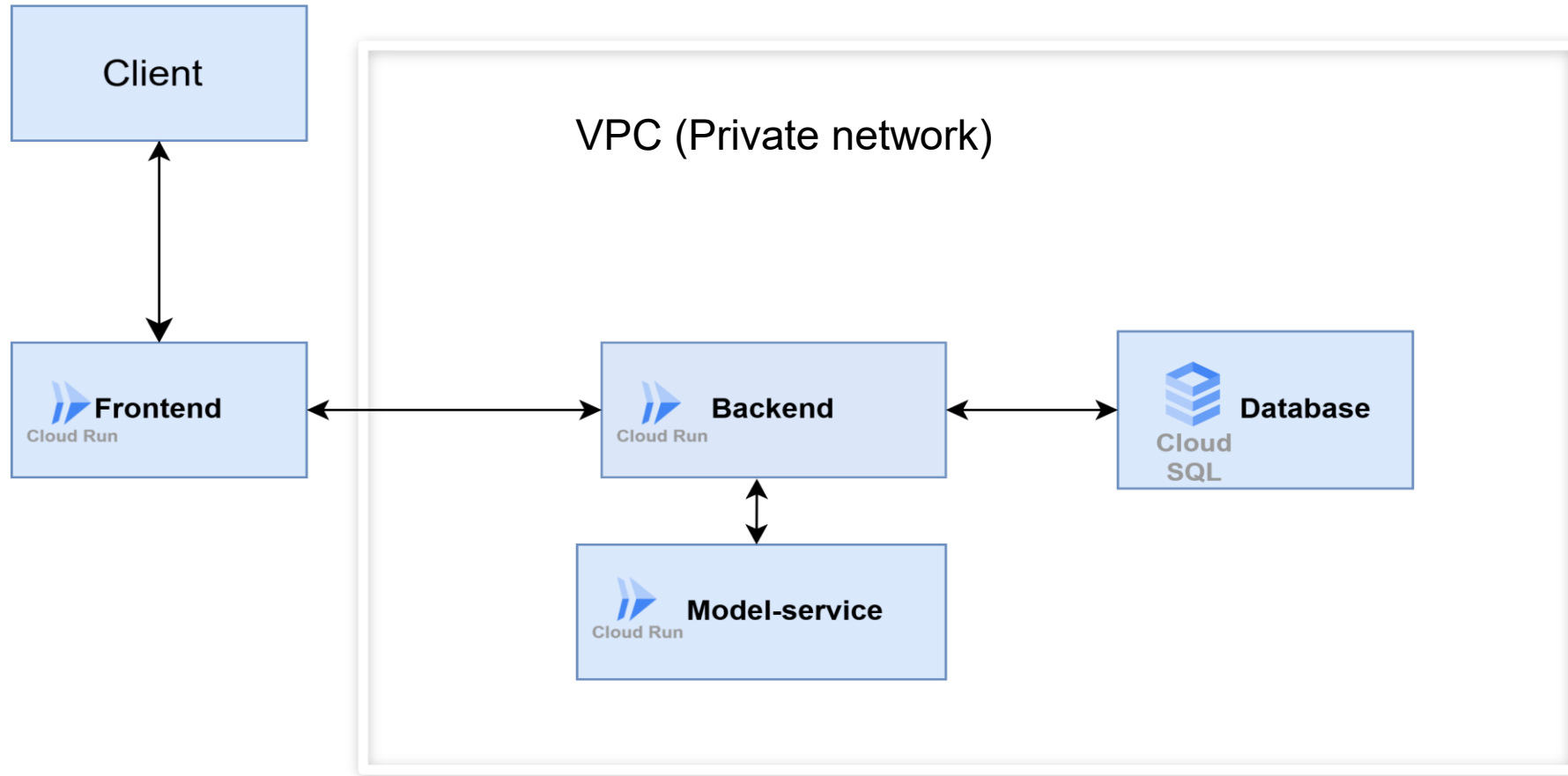


Why ruff?

Ruff is an extremely fast Python linter and code formatter.

We used Ruff to automatically check code quality and catch issues before deployment.

SERVERLESS DEPLOYMENT ON GOOGLE CLOUD RUN



FRONTEND



Quick Demo of the application:



<https://cinematch-frontend-4v2wwdcp7a-ew.a.run.app/>

GITHUB



<https://github.com/HemaNaidu91/tsm-mach-le-datagoup3>